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Authors

Dr. WILSON CASTRO

Affiliations
• Universidad Nacional de Frontera, Sullana, Piura, Peru

Dr. HIMER AVILA-GEORGE

 [ORCID](https://orcid.org/0000-0001-8578-0170)
<https://orcid.org/0000-0001-8578-0170>

Affiliations
• Departamento de Ciencias Computacionales e Ingenierías,
Universidad de Guadalajara. Ameca, Jalisco, Mexico

Mr. WILLIAM NAURAY

Affiliations
• Dirección de Estudios e Investigación, Servicio Nacional Forestal y
de Fauna Silvestre, Lima, Peru

Dr. ROENFI GUERRA

Affiliations
• Universidad Nacional del Altiplano, Puno, Peru

Mr. JORGE RODRIGUEZ

Affiliations
• Universidad Nacional de Piura, Piura, Peru

Mr. JORGE CASTRO

Affiliations
• Universidad de Cuyo, Mendoza, Argentina

Dr. MIGUEL DE-LA-TORRE
Corresponding Author
Submitting Author

- Affiliations**
- Departamento de Ciencias Computacionales e Ingenierías,
Universidad de Guadalajara. Ameca, Jalisco, Mexico

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Deep Classification of Algarrobo Trees in Seasonally Dry Forests of Peru using Aerial Imagery

WILSON CASTRO¹, (Senior Member, IEEE), HIMER AVILA-GEORGE², WILLIAM NAURAY³, ROENFI GUERRA⁴, JORGE RODRIGUEZ⁵, JORGE CASTRO⁶, MIGUEL DE-LA-TORRE²

¹Universidad Nacional de Frontera, Sullana, Piura, Peru.

²Departamento de Ciencias Computacionales e Ingenierías, Universidad de Guadalajara. Ameca, Jalisco, Mexico.

³Dirección de Estudios e Investigación, Servicio Nacional Forestal y de Fauna Silvestre, Lima, Peru.

⁴Universidad Nacional del Altiplano, Puno, Peru.

⁵Universidad Nacional de Piura, Piura, Peru.

⁶Universidad de Cuyo, Mendoza, Argentina.

Corresponding author: Miguel De-la-Torres (e-mail: miguel.dgomora@academicos.udg.mx).

ABSTRACT Seasonally dry forests present special requirements to face deforestation and desertification, with particular interest in protected species of trees. Efforts to defend protected flora involve the maintenance of an inventory of seed-producer trees (*plus trees*), with the aim of making informed decisions in specific zones. In this paper, a methodology is proposed for the automated classification of *plus* algarrobo trees (*Neltuma pallida*), based on RGB aerial imagery and deep learning classifiers. As a proof of concept, three *state-of-the-art* approaches were evaluated in selected zones of interest, and a new application-specific architecture called AlgarroboNet was proposed for efficient classification of *plus* algarrobo trees. First, a database with the location and phenological characteristics of algarrobo trees was provided by the Peru National Forest Service. *Plus* trees were measured to obtain their morphometric characteristics, and an *in-situ* evaluation was conducted, capturing aerial images of *plus* and *non-plus* trees. After correction and segmentation, a balanced database was prepared to evaluate the four deep classifiers. The performance of each approach was summarized using both accuracy and F_1 -measure, following a *hold-out* validation strategy with 30 trials. The results reveal that *state-of-the-art* approaches show a F_1 -measure that ranges between 0.92 and 0.99 among the models, presenting a trade-off between computational resources and accuracy. In the analysis, the proposed methodology for massive monitoring and characterization of *plus* algarrobo trees was found to be suitable for implementation, aiming to maintain an up-to-date inventory and enforce reforestation programs.

INDEX TERMS Deep classification, aerial imagery, algarrobo trees, dry forests, AlgarroboNet.

I. INTRODUCTION

IDENTIFICATION of tree species, and particularly those that are seed producers within the same species, is a fundamental task in remote sensing to address ecological research [1]. The preservation of forest resources is crucial, as they support 80% of amphibian species, 75% of bird species, and 68% of mammal species on Earth [2]. However, monitoring protected areas can become challenging due to the extension of the terrain and limited accessibility. In fact, forests cover over 31 % of the global land surface, with more than 4 million hectares that are affected by various problems related to illegal logging, wildfires, plagues, and diseases,

among others. In particular, dry tropical forests are subject to special attention due to their location in tropical regions with pronounced seasonality in the distribution of precipitation, resulting in several months of drought [3]. These changing conditions favor dry tropical forests to develop a significant level of biodiversity, with 12 different floristic groups in Central and South America, and between 100 and 1000 tree varieties according to their position relative to the equator [4]. Furthermore, tropical dry forests are home to various wild animals such as big cats, monkeys, parrots, rodents, wild birds, sloths, black howlers, monkeys, foxes, raccoons, badgers, lynxes, pumas, etc. [5].

Despite the ongoing decline of forested areas, between 2011 and 2020 forests were able to sequester more carbon dioxide than they released. According to the Food and Agriculture Organization of the United Nations, this result can be attributed in part to efforts such as reforestation and enhanced management practices [6]. The expansion of forested areas has been achieved through reforestation efforts, resulting in a 123 million hectare increase since 1990 to a current total of 294 million hectares [2]. However, various factors, including environmental constraints and human activities, produce deforestation and degradation and continue to limit the success of these reforestation initiatives [7]. As an example, in the last two decades, there has been a significant amount of deforestation and a lack of vegetation in various regions of Peru, the most noticeable impact observed near the Amazon region. However, this deforestation is not uniform and seriously affects relevant ecosystems such as dry forests, where protected species of trees such as algarrobo [8].

With regard to the main topic of this paper, the algarrobo tree (*Neltuma pallida*), which is a native species in the arid and semi-arid regions of South America, is a deciduous tree that has the ability to thrive in dry ecosystems. It is known for its resilience against challenging environments, including high salinity levels, drought conditions, and high temperatures [9]. The consumption of this plant as food, along with its high fiber and nutrient content, makes it economically important. In addition, it serves as a valuable source of honey and wood and plays a relevant role in improving soil and environmental conditions [10], [11].

However, the geographical distribution of the algarrobo trees is influenced by climatic occurrences like “El Niño”, resulting in significant variations between the arid and rainy seasons. The adaptation capacity of the algarrobo trees allows the species to thrive both during drought years and in the aftermath of floods. These fluctuations have been shown to affect the survival and growth of algarrobo trees, as documented in [9]. In addition, despite its adaptability, the spread of the algarrobo tree along its native habitat faces threats from anthropogenic pressure and habitat conversion, resulting in loss of income for communities [10]. For example, some human activities are important to understand the growth behavior in specific geographic zones, including cultural, environmental, and economic factors, aesthetic, spiritual value, etc. [12]. To address the constantly changing environment, it is essential to implement a robust reforestation policy. This policy should include maintaining a georeferenced inventory of algarrobo tree populations. Furthermore, a key activity for safeguarding the reproduction conditions of the species involves identifying *plus trees* and taking measures to protect them [12].

In practice, a standardized method commonly used to identify algarrobo *plus trees* is based on characteristics such as crown size, percentage of healthy stems, and measurement of trunk diameter and height. These criteria are flexible and can be modified according to changes in forest management objectives, climate, social expectations, and adaptability to

the biophysical conditions of the study area [13]. The collection of these data for inventory purposes implies a laborious *in situ* process that involves the selection and measurement of a forest region, collecting the composition, density, diameter, height and other parameters of the species to estimate the characteristics of the forest as a whole [14].

An alternative to *in-situ* evaluation is the use of remote sensing imagery, which allows obtaining information about the Earth’s surface without direct physical contact and is carried out through the use of remote devices and sensors, such as satellites, airplanes, drones or ground sensors [15], [16]. In this way, spatial and temporal information can be combined to establish measurements related to the state of the crop in a non-destructive manner, facilitating in this way sustainable crop management [15], [17]. In recent developments related to remote sensing, the large volume of spectral data retrieved from remote devices has been automatically analyzed with computational tools. The methods used to analyze acquired spectral images usually include machine learning (ML) and image processing techniques, designed to extract useful information that provides insight to make informed decisions [17]–[19]. Among the most relevant ML techniques, deep learning approaches take advantage of complex architectures that provide specialized modeling strengths in high-dimensional classification problems. The most relevant deep approaches for remote sensing imagery employ convolutional neural networks (CNN), capable of extracting information and self-organizing, shifting the extraction process of features of interest, and focusing on the learning system [20], [21]. The category of CNN approaches includes numerous algorithms that are widely used in image analysis. In particular, in the forest sector, it has been shown to be highly effective in identifying and segmenting trees from other vegetation using images captured with unmanned aerial vehicles (UAV) [19], [22]. In general, CNN training approaches involve the construction of architectures with several parameters, distributed across many processing layers. Resulting models are commonly complex in terms of memory resources, with the consequent high computational cost.

In this paper, four main contributions are reported as a result of the research process. First, a methodology is proposed to use CNN-based deep learning algorithms and aerial images to automatically classify *plus* and *non-plus* algarrobo trees. Second, the proposed methodology was applied to three zones of interest that correspond to the area of study, and the supporting results are reported as a relevant case of study. Third, a new compact and application-specific architecture is proposed to classify aerial images of algarrobo trees, the so-called *AlgarroboNet*. Finally, as a result of the application of the proposed methodology, a deep learning technique comparison was performed, and the results were reported and discussed.

The remainder of the paper is organized as follows. Section II provides details on the area of study, the location, and general weather conditions. Section III describes the

TABLE 1: Districts of the northern coast of Peru with a high probability of plus trees. Central coordinates for each district are expressed using WGS84.

Department	Province	District	Latitude	Longitude
Tumbes	Contra almirante Villar	Zorritos	-3.6808	-80.6825
	Tumbes	San Jacinto	-3.6917	-80.4686
Piura	Morropon	Chulucanas	-5.0921	-80.1628
		La Matanza	-5.2833	-80.5833
	Piura	Veintiseis de Octubre	-5.1866	-80.6540
	Sechura	Sechura	-5.5569	-80.8228
	Sullana	Marcavelica	-4.8778	-80.7183
Lambayeque	Lambayeque	Olmos	-5.9850	-79.7433

proposed methodology, as well as the new *AlgarroboNet* approach designed for the application. Section IV shows the relevant results and discussion. Finally, Section V presents the conclusions and projects the future research directions.

II. AREA OF STUDY

The area of study includes eight districts, spanning across three out of the twenty-four departments in which the territory of Peru is divided. The eight districts under study are listed in Table 1 and reveal different scenarios to analyze in aerial imagery.

Fig. 1 presents the location of the zones of interest on a global map and according to the political division of Peru. In the first place, two districts were selected at the Department of Tumbes, which is the smallest and northernmost region of Peru, known for its coastal and tropical landscape with rich biodiversity that represents variations in the scenario. The Department of Tumbes is bordered by Ecuador to the north, the Pacific Ocean to the west, and the Department of Piura to the south. The districts selected in this study include the coastal Zorritos, belonging to the province *Contra almirante Villar*; and San Jacinto, belonging to the province *Tumbes*, with a mix of agricultural and residential areas. In the second place, five districts were selected from the Department of Piura, which covers a coastal region in northwestern Peru. It is bordered by the Pacific Ocean to the west, the Department of Tumbes to the north, the Department of Lambayeque to the south, and Ecuador to the east. The districts of the Department of Piura include Chulucanas and La Matanza in the province of *Morropon*; Veintiseis de Octubre in the province of *Piura*; Sechura, in the province of the same name; and Marcavelica, in the province of *Sullana*. Finally, the district of Olmos was selected from the Department of Lambayeque, in the province of the same name. The district of Olmos is known for its large agricultural fields and irrigation systems that favor the growth of crops of interest. Thus, the landscape is modified based on agriculture requirements, and special policies are of interest to preserve protected species.

The criteria followed to select the study area included the inclusion of the zone covered in the SERFOR studies developed in 2021 and 2022, as well as information related to seasonally dry forest ecosystems associated with *Prosopis* [23]. At the same time, satellite images (Google Earth)

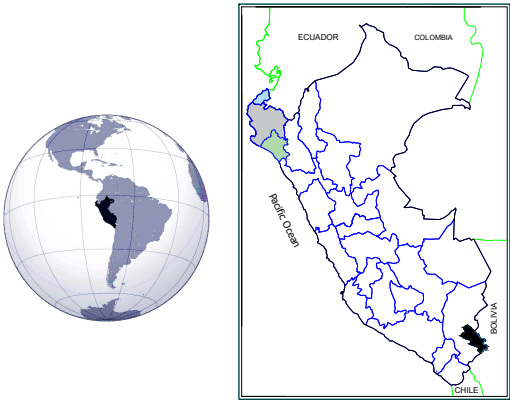


FIGURE 1: Location of the Departments of Peru, where the aerial images were captured. These Departments were included in the studies developed by SERFOR in 2021 and 2022 and present seasonally dry forest ecosystems associated with algarrobo trees

were verified to be freely available and capture zones were chosen on the northern coast of Peru, where visual inspection revealed a high probability of finding *plus trees*.

III. METHODOLOGY

The methodology proposed for the detection and classification of algarrobo trees in aerial imagery is summarized in Fig. 2. In general, four phases are established according to the practice. Each phase is divided into different, more simple steps. For a start, an *in-situ* evaluation was conducted to manually classify *plus* and *non-plus* algarrobo trees under the supervision of forest experts and establish the *ground truth* of selected trees. Concurrently, aerial images were acquired, georeferenced, and categorized for further processing. The resulting aerial images were manually corrected, and the tree crowns were segmented to be integrated into the image database. Finally, the crowns of the trees were characterized to obtain their morphometric parameters, and deep models were trained and evaluated to obtain performance metrics for comparison. The main steps of the proposed methodology are

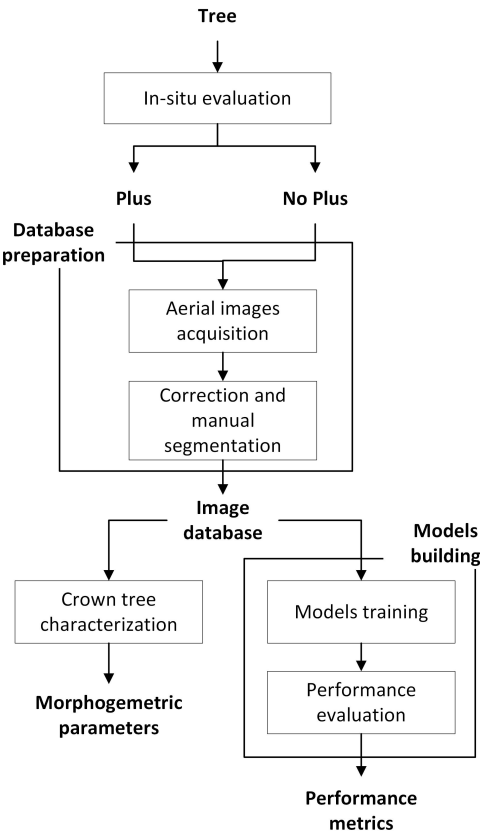


FIGURE 2: Proposed methodology for the evaluation of deep approaches to classify *plus* and *non-plus* algarrobo trees.

shown in Fig. 2 and detailed in the following paragraphs.

Computer-aided analysis of aerial images of algarrobo trees was supported with functions and graphical user interfaces implemented in Matlab 2023^a (Mathworks co). The `trainNetwork` function of the *Deep Learning Toolbox*TM was used in all cases to train all the deep approaches. Likewise, the representation of the study locations was implemented using AutoCAD (Autodesk co.) version 2024. The computer used for simulations was an Intel(R) Xeon(R) Gold 5218, with two processors running at 2.30GH, 96MB RAM, running a 64 bit Windows 10 Pro.

A. IN-SITU EVALUATION

As the first phase with a single step, the *in-situ* evaluation samples were retrieved and categorized under the supervision of forestry experts. The category of *plus* trees included those characterized as phenotypically outstanding concerning the individuals of a population, with one or several characteristics of economic interest according to the proposal of [24]. Algarrobo trees that do not fit this characterization were classified as *non-plus* trees. In this manner, the training dataset was assembled with segmented and labeled images.

The brigade in charge of data collection was displaced to the previously selected locations. Once there, members of the

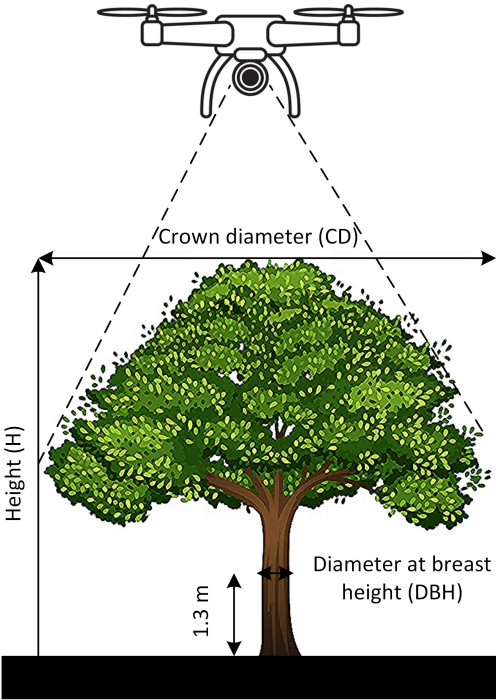


FIGURE 3: Illustration of the *in-situ* measurement of the morphometric parameters and capture of aerial imagery of the trees employed in the study. The process precedes the manual classification in *plus* and *non-plus* trees

brigade applied the *in situ* collection protocol by measuring samples and obtaining aerial images, as illustrated in Fig. 3.

The minimum measurements to consider a specimen as a *plus* tree are listed in Table 2. The *in situ* measurement process was repeated in different algarrobo trees until 500 *plus* trees were registered. The same condition was applied to *non-plus* trees to build a database of at least 1000 measurements.

TABLE 2: Lower limits in the measurement in the *in-situ* evaluation of algarrobo trees, in order to be categorized as *plus*

Measurement	Notation	Lower limit
Diameter at Breast Height	<i>DBH</i>	0.45 m
Height	<i>H</i>	13 m
Crown Diameter	<i>CD</i>	10 m

Each of the independent trees was georeferenced using a GPS (GPSMAP 64sx, Garmin co.), and the position was stored using the standard notation Urchin Tracking Module (UTM).

B. DATABASE PREPARATION

The information retrieved from the *in-situ* evaluation was captured and prepared for compatibility with existing reference models. Thus, coordinates, morphometric parameters, and aerial images were organized in a dedicated database for further analysis and model building. Along this line, the

database preparation phase involves the two steps detailed below.

1) Aerial images acquisition

Starting from the geolocation of the *plus* and *non-plus* trees, the flight plans were designed and loaded onto a quadcopter drone (Phantom 4 multispectral - DJI, China). The images captured with the drone consisted of five spectral bands corresponding to the blue, green, red, red edge, and near-infrared colors, all with a resolution of 2 MP. The database included five hundred images for each of the classes, and according to the protocol, the capture time ranged between 9:00 and 15:00 hours, with a fixed flight height of 70 meters above ground level. The resulting images were stored in the standard format .TIFF and organized by area, date, and class. Finally, red, green, and blue bands were assembled in a .jpg format to delimit the color information and standardize compatibility with known deep learning models.

2) Correction and manual segmentation

The RGB were corrected to reduce the errors that resulted from displacement in the position of the lenses. Fig. 4(a) shows an example of the resulting corrected RGB images. Then, the operator manually selected the area covered by the crown of each individual tree and generated the files with reference regions of interest (ROI), see Fig. 4(b), following the procedure reported in the literature [25], [26]. The reference ROI files were created, preserving the pixel dimensions of the original files, and stored in cloud storage systems for further analysis and processing.

C. CROWN TREE CHARACTERIZATION

The image-based morphometric characterization of the tree crowns considers the manually segmented ROIs, extracted according to the procedure proposed in [27]–[29]. The eight morphometric parameters obtained from the isolated ROIs include the minor and major axis, perimeter, and area, as illustrated in Fig. 4, and summarized in Table 3.

The basic morphometric parameters shown in Fig. 4(c) were estimated considering the dimension of the pixels related to the distance of the drone from the crown of the tree. The eight morphometric parameters are listed in Table 3.

As shown in Table 3, the crown area is estimated by multiplying the number of pixels within the ROI by the area of a pixel. The side length and area of a pixel in the crown are approximated by considering the dimensions of known objects and the distance from the camera to the crown.

D. MODELS BUILDING

Numerous CNN-based models have been proposed for image classification, providing different complexities and accuracies for distinct classification problems. These models present a trade-off between complexity and accuracy, influenced by architecture, activation functions, and training strategy [30]. Below, the architectures of three reference

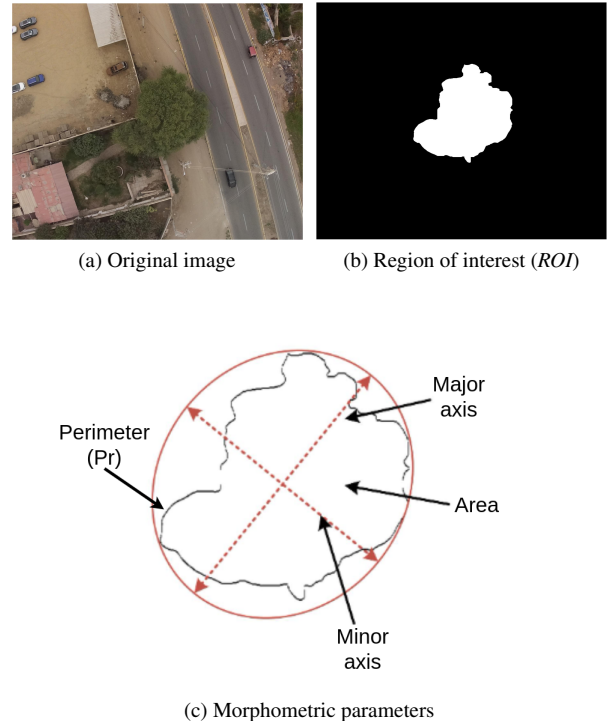


FIGURE 4: Results of the database preparation: (a) RGB original image, (b) manually segmented ROI, and the morphometric parameters of the crown, obtained after image analysis

models are briefly described, and the proposed *AlgarroboNet* is explained in depth.

1) Models training

Input training and test images were prepared to fit the requirements of the reference and proposed models. The images in the dataset were resized according to the proposals of [31], to be used as input in the CNN training, so the images in the dataset were resized to $[224 \times 224]$ and $[227 \times 227]$ pixels per channel using the *bi-cubic* interpolation method.

Next, three deep classifiers based on known convolutional neural networks were implemented in a manner analogous to [32], as detailed below, and the proposed architecture as shown in Subsection D.2.

- AlexNet. Proposed by Alex Krizhevsky in 2012, AlexNet is one of the most well-known convolutional neural networks (CNNs) in the image processing community. The architecture consists of five convolutional layers followed by three fully connected layers, totaling around 62.3 million parameters [31], [33], [34]. The network employs the Rectified Linear Unit (ReLU) activation function in both its convolutional and fully connected layers. ReLU is effective in addressing the problem of vanishing gradients, which contributes to the successful training of deep neural networks. In terms of training parameters, AlexNet uses an initial learning rate of 0.01, which is reduced by a factor of 10 when

TABLE 3: Morphometric parameters estimated on the region of interest retrieved from the segmented aerial images (see Fig. 4). The *numROI Pixels* is the raw counting of white pixels in the segmented ROI, the *pixelArea* and *pixelSide* are averaged based on *in-situ* measurements of *in-situ* measurement of the crown, and *numPr Pixels* is the raw counting of pixels at the border.

Parameter	Description	Estimation
Area (A)	The crown area seen from above	$numROI Pixels \times pixelArea$
Equivalent diameter (Ed)	Average diameter of the surrounding ellipse	$Average\ diameter$
Major axis length (ML)	The greatest axis of the surrounding ellipse	$Largest\ axis$
Minor axis length (mL)	The smallest axis of the surrounding ellipse	$Smallest\ axis$
Perimeter (Pr)	The length of the surrounding line of the crown	$numPr Pixels \times pixelSize$
Roundness (Rd)	Estimate of the circularity of the crown	$Rd = \frac{4\pi A}{Pr^2}$
Elongation (El)	The ratio between ML and mL	$El = \frac{ML}{mL}$
Compactness (Cp)	A measure of the denseness of the branches/leaves	$Cp = \sqrt{\frac{4A}{\pi}}$

the validation error stagnates. It also employs a weight decay of 0.0005 for regularization and a momentum of 0.9 to enhance gradient descent. The model is trained with a batch size of 128 over 90 epochs, optimizing its performance on the ImageNet dataset. These parameters are crucial to achieving the high performance of AlexNet in image classification tasks [34].

- DenseNet-201.- DenseNet-201 is a CNN architecture composed of three types of blocks: convolution blocks, dense blocks, and transition blocks that connect two contiguous dense blocks. This structure results in a model with 40 million parameters [32], [35]. According to [36], DenseNet-201 mainly uses the Rectified Linear Unit (ReLU) activation function in its convolutional layers to address the vanishing gradient problem and accelerate training. Additionally, alternative activation functions, such as the parametric rectified linear unit (PReLU) and the exponential linear unit (ELU) can be integrated to enhance adaptability and performance in specific contexts. DenseNet-201 adjusts a number of important parameters during training, such as the learning rate, the number of training epochs, the batch size, the momentum for gradient descent optimization, and the weight decay for regularization. These parameters are essential for fine-tuning the DenseNet-201 architecture.
- GoogLeNet.- Also known as Inception v1, GoogLeNet is a CNN architecture developed by researchers at Google, consisting of 22 layers [37]. It gained prominence by winning the ImageNet Large-Scale Visual Recognition Challenge (ILSVRC) in 2014 and has since been applied to various tasks such as object detection, scene recognition [38], and remote sensing [39], [40]. GoogLeNet is notable for its innovative approach to network architecture, significantly improving the efficiency and performance of deep learning models for image classification; its architecture employs a diverse set of activation functions to enhance model performance. The use of Rectified Linear Unit (ReLU) in convolutional

layers effectively addresses the vanishing gradient problem. Additionally, the integration of parametric rectified linear units (PReLU) and exponential linear units (ELU) provides improved adaptability and accuracy. Regarding training parameters, GoogLeNet incorporates key factors such as learning rate adjustment, weight decay for regularization, momentum for gradient descent optimization, batch size determination, and the specification of the number of training epochs. These parameters are crucial in optimizing the GoogLeNet architecture for efficient image classification tasks, demonstrating its effectiveness in deep learning applications [37].

2) Algarrobonet

The proposed architecture to classify *plus* and *non-plus* algarrobo trees (AlgarroboNet) is shown in Fig. 5. The process is distributed in a total of 18 layers as follows. First, inspired by the AlexNet architecture, the input image is resized to a $227 \times 227 \times 3$ matrix, and a $5 \times 5 \times 3$ convolution layer constitutes the first filtering of the network. Then, a batch normalization is applied, followed by a ReLU activation function, a MaxPooling layer to reduce the spatial dimensions, and a second convolution layer. The resulting structure is further processed by a second batch normalization layer, a second ReLU activation function, and a MaxPooling layer followed by a third convolution layer to obtain a reduced representation of input data. A third batch normalization layer is then applied, followed by a third ReLU activation function and a third MaxPooling layer. Finally, a fourth convolution layer is applied, followed by a batch normalization and ReLU activation function that feeds a fully connected layer. A Softmax activation function provides the feed for the 2-neurons output layer, predicting either *plus* or *non-plus* classes. In summary, the eighteen layers represent a 4-steps sequential process that involves a convolution layer, assembled with batch normalization, a ReLU activation function, and a MaxPooling layer. The last of the four steps substitutes MaxPooling by a fully connected layer that merges all synaptic responses, summarizing the total activation of the network

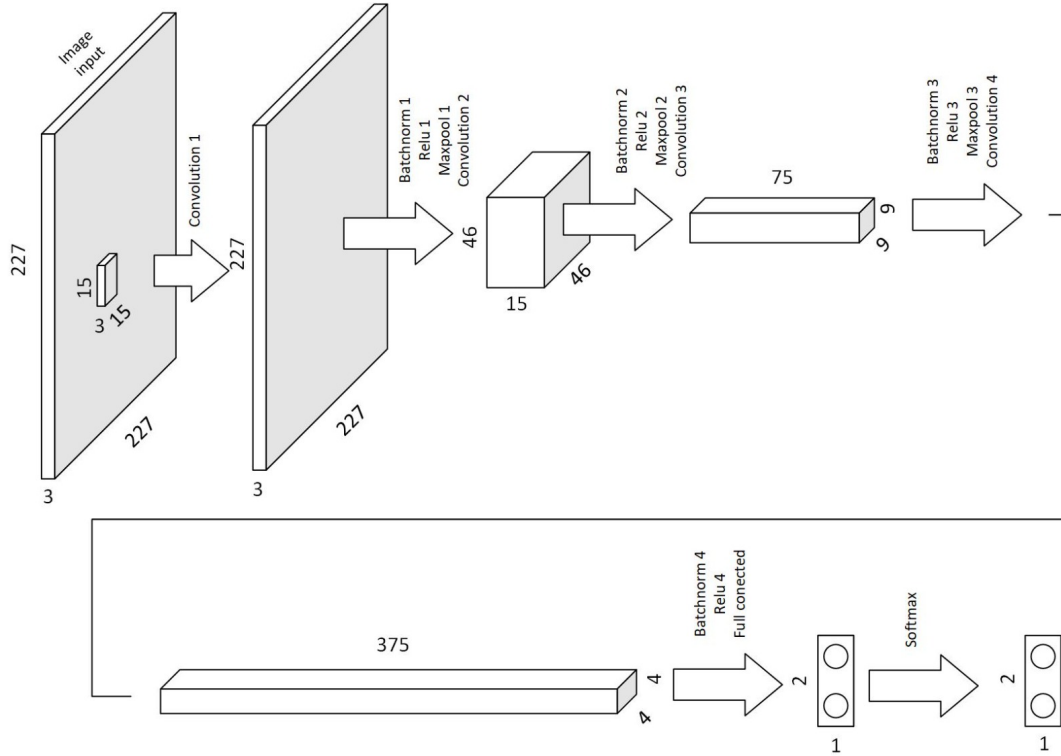


FIGURE 5: Architecture of the proposed *AlgarroboNet* convolutional network to classify *plus* and *non-plus* algarrobo trees. The architecture consists of eighteen layers that produce a binary decision out of a $227 \times 227 \times 3$, color (*RGB*) input image

into two values that feed the Softmax activation function and produce a *plus/non-plus* class label.

In general, the *AlgarroboNet* architecture is made up of four convolutional layers, four batch normalizations, four ReLU activation functions, three MaxPooling layers, a SoftMax activation function, and the 2-neurons output layer. Training the *AlgarroboNet* network requires determining a total of 6 606 806 trainable parameters, distributed along the 18 layers.

3) Performance evaluation

Training data was randomly split and organized into 30 different training/test subsets, following a *hold-out* evaluation strategy with 30 independent trials. The performance for each trial was first computed in terms of a 2-class confusion matrix (CM), summarizing the amount of true positives (*TP*), true negatives (*TN*), false positives (*FP*) and false negatives (*FN*). The CM organizes the true and predicted classes of the elements of a sample in a square $[n \times n]$ array, where $n = 2$ for *plus* and *non-plus* classes. Then, other measures derived were calculated, following standard criteria [41] that have been used in multiple applications [32], [42]. Comparison of the performance for each model was organized in terms of accuracy and F_1 -measure, determined using Eq. 1 and Eq. 2.

$$Accuracy = \frac{TP + TN}{TP + FP + FN + TN} \quad (1)$$

$$F_1measure = 2 \left(\frac{PR \times RC}{PR + RC} \right) \quad (2)$$

where *Accuracy* is the rate of well classified instances among the total number of samples in test, and the F_1 -measure is the harmonic mean between PR (precision) and RC (recall). Additionally, TP + FN are the total positive cases, and TN + FP are the number of negative samples in the test set.

IV. RESULTS AND DISCUSSION

The proposed methodology was validated using the measurements and images retrieved *in-situ*, the three *state-of-the-art* approaches (*AlexNet*, *DenseNet*, and *GoogLeNet*), as well as the proposed *AlgarroboNet* model. As a complementary result, the four approaches are compared using the methodology, and the results are reported throughout this section. Similarly, some limitations and opportunities for improvement are discussed.

A. IN-SITU EVALUATION

The expedition to evaluate and retrieve aerial images of the Algarrobo trees was guided by the *SERFOR* report, which includes the location of the previously registered trees. As expected, in this new field exploration some changes in the landscape were detected. For instance, the Algarrobo trees that were found were significantly less numerous than those

reported by *SERFOR*. The trees found *in-situ* with coincidence in the report were mapped in the new dataset used for the validation of the methodology.

According to the observations in the capture locations, the change in the number of Algarrobo trees corresponds to a diversity of situations. Some of the reasons for this reduction are related to the loss of particular samples due to diseases, weather conditions, or the increase in deforestation to consume and trade Algarrobo charcoal. In fact, the relationship between the reduction of Algarrobo tree populations and factors such as urban growth, agricultural expansion, and wood harvesting has been found to significantly affect the population of the specie [43].

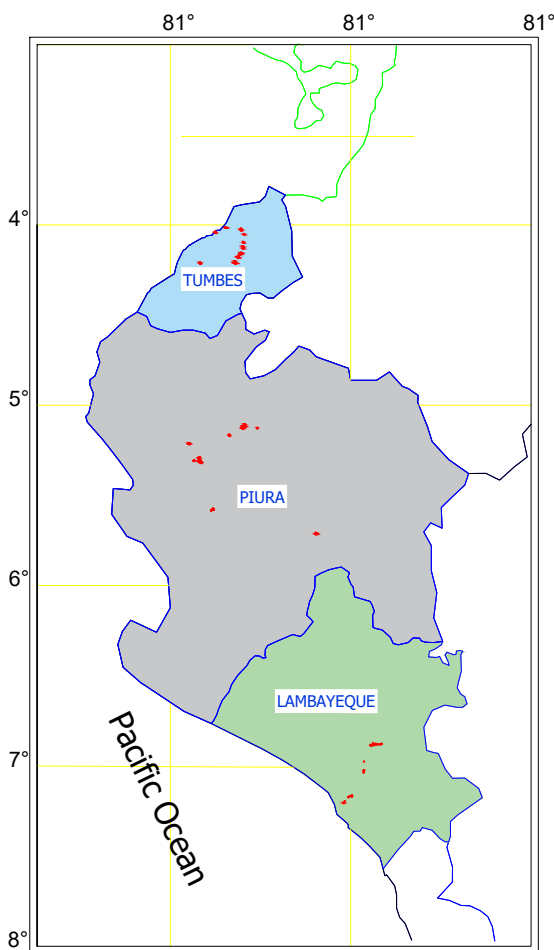


FIGURE 6: Location of the Algarrobo *plus* trees found during the *in-situ* evaluation. Both manual measurements and aerial imagery were recorded for further processing.

According to the results of the *in-situ* evaluation, the final dataset after depuration of unuseful data resulted in 500 samples of *plus* trees, and 495 samples of *not plus* trees. Fig. 6 depicts with red marks the position of the Algarrobo *plus* specimens that were registered for this study, and are relevant to preserve and reproduce the specie. All the specimens registered in the dataset were validated with the *SERFOR*'s report [44].

B. CROWN TREE CHARACTERISTICS

Fig. 7 shows the histograms corresponding to the distribution of the morphological characteristics of the crowns of Algarrobo trees registered in the dataset. This representation follows those reported by other authors that have used morphological characteristics to evaluate UAV with RGB sensors. As examples, morphological characteristics were used as input values for models trained to identify pineapple crowns [45], detect sorghum heads [46], and estimate the cotton mass [47].

According to Fig. 7(a), the area of the crown of the Algarrobo *plus* trees is consistently higher than the area corresponding to *non-plus* trees, although a small overlap is present close to 100 m^2 , which could be a good guess for a separation threshold between classes. It is also evident from Fig. 7(a) that a few specimens surpass 500 m^2 , that correspond to trees of exceptional size. A similar trend is observed in Fig. 7(b), showing that both areas and perimeters for both classes show a positive bias to the right. However, the shape of the perimeter distribution for the *non plus* trees is closer to a Gauss bell, with a measure of central tendency close to 30 m , against the 60 m that correspond to the *plus* trees. Similarly, a small overlap between class distributions is present, which is higher at 40 m . Following the rest of the images in Fig. 7, a similar trend is observed in Fig. 7(c), (d), and (e) that represent the distribution of the major and minor axes, as well as the equivalent diameter, respectively. In these three histograms, the separation between distributions of *plus* and *non-plus* trees is clear, with a small overlap close to 15 m for the major axis, 10 m for the minor axis, and 10 m for the equivalent diameter. The shape of the distributions of major and minor axes, as well as the equivalent diameter, resembles the normal distribution with different measures of central tendency and dispersion. In these five cases (Fig. 7(a-e)), the measurement of the morphological characteristics for *plus* trees are higher than those corresponding to *non-plus* Algarrobo trees. In contrast, the roundness distributions shown in Fig. 7(f) reveals that *non-plus* trees present a slightly higher roundness than the *plus* trees, with a high overlap.

C. MODELS BUILDING

The first three pre-trained networks employed as reference approaches (*AlexNet*, *DenseNet*, and *GoogleNet*), take advantage of transfer learning in a pre-trained architecture. In contrast, the proposed architecture called *AlgarroboNet* was trained from scratch, considering the well-delimited classification task to discriminate among *plus* and *non-plus* algarrobo trees. Training was performed over 160 iterations to evaluate the highest possible performance on each particular trial. Fig. 8 presents the evolution of the accuracy, from 0 to 160 iterations. In all cases, the accuracy achieved during training reached a plateau that denotes stability of the model, before reaching one hundred iterations. In the case of the proposed *AlgarroboNet*, the maximum accuracy was completed at exactly 60 iterations. Similarly, the precision of the pre-trained models was stable, with slight differences,

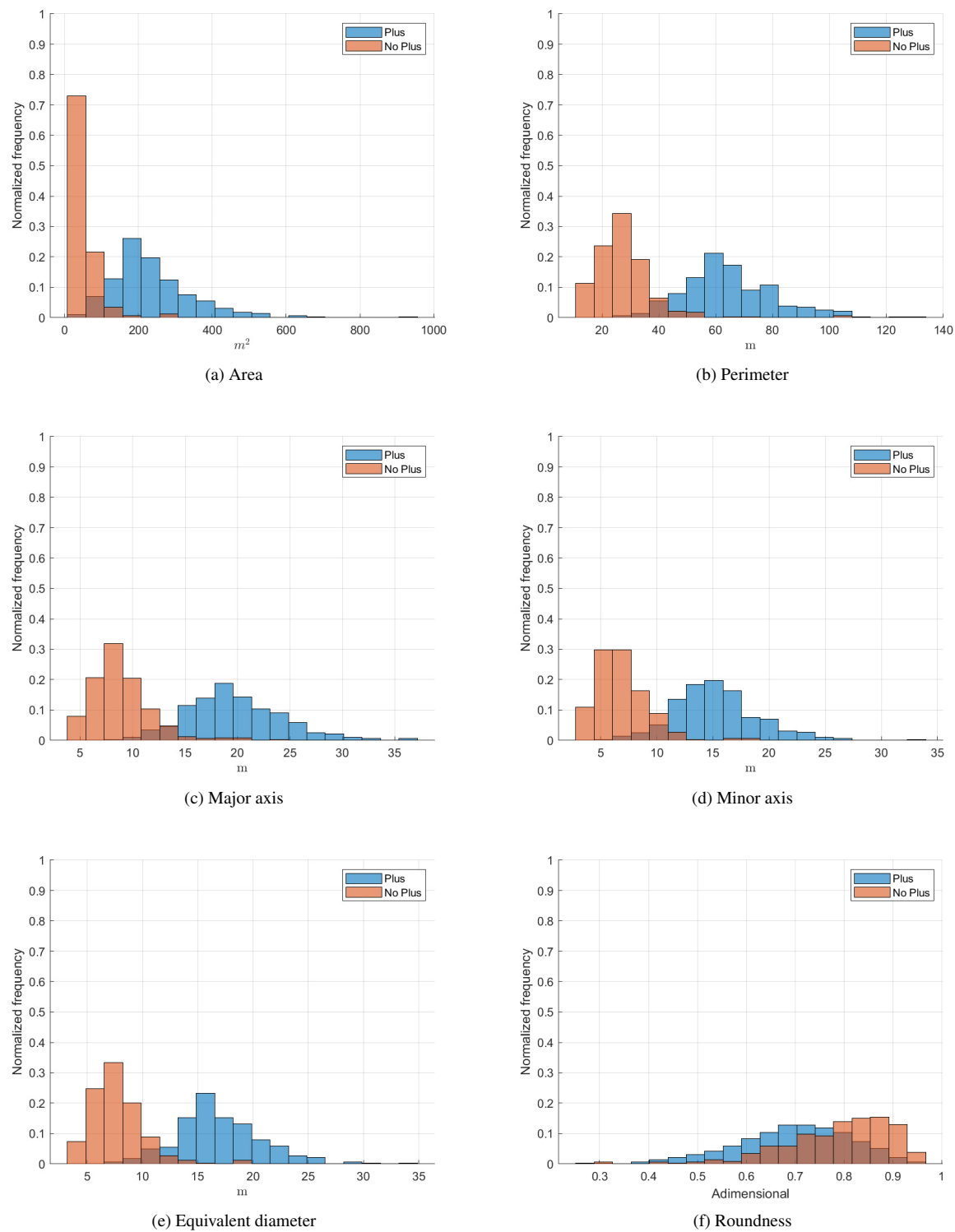


FIGURE 7: Normalized distribution of the morphometric parameters for the *plus* (blue), and *non-plus* (orange) algarrobo trees employed in the study.

after exceeding 40 iterations, which is related to the knowledge previously learned by the pre-trained approaches. These results correspond to the results reported in literature [32]. On the other hand, the lowest accuracy value was shown

by GoogleNet, which stabilized around 95 % at around 40 iterations.

In general, the accuracy for all approaches usually increases with the number of iterations. However, after a certain

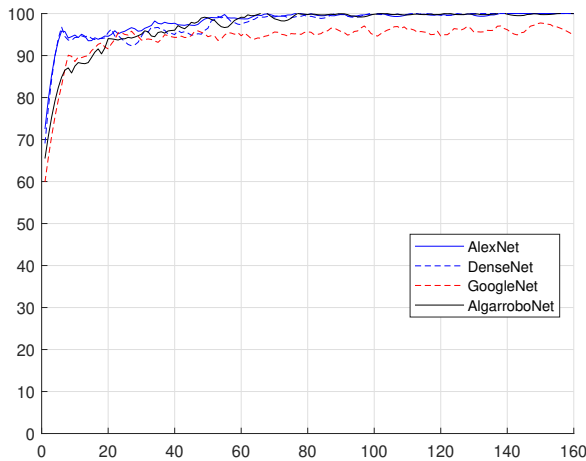
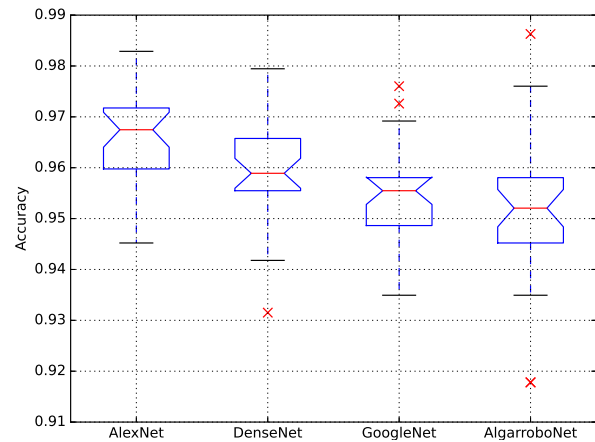


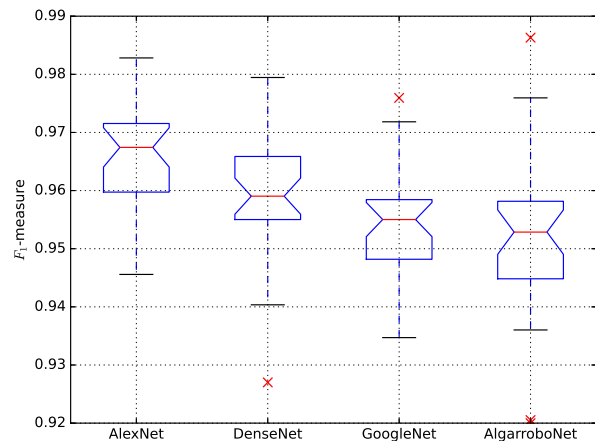
FIGURE 8: Comparison of the performance of *CNN* models for the discrimination of plus and *non-plus* trees

number of epochs, the network’s recognition accuracy may stop growing or even decrease [48]; therefore, to obtain a maximum accuracy, it is required to monitor and select the number of iterations during the model training. This upper limit is reached at the beginning of the training of the proposed *AlgarroboNet*, a network that was initially more accurate than *GoogleNet*, but showed greater and evident instability throughout the training, reaching close to 100% accuracy at around 65 iterations, but progressively decreasing and rising again to its maximum at around 90 iterations. This effect of “slowdown” and “speedup” was reported in a model described in [49] and [50] where the number of training epochs depends on the architecture of the network, which is directly proportional to its depth.

Following the evaluation of the proposed methodology, Fig. 9 shows the box plots of the accuracy and F_1 -measure corresponding to the four models that were evaluated. The evaluation followed a 30-times hold-out validation strategy, with 70% training and 30% test samples, randomizing each time. In general, no significant differences were found between the accuracy distributions and the measure F_1 for the four models. However, *AlexNet* showed a higher median accuracy and F_1 -measure, and the proposed *AlgarroboNet* showed a slightly lower accuracy and F_1 -measure compared to *GoogleNet*. The performance achieved by the *AlgarroboNet* can be easily explained by the lack of prior knowledge, compared to the other models: no transfer learning was done for the *AlgarroboNet*. In fact, it has been shown that using transfer learning, researchers such as [51], [52], [53] achieved precisions above 88%; this is based on the adaptive capacity of these networks previously trained with massive amounts of data [54]. However, it can also be seen in Fig. 9 that the model that achieved the highest performance in terms of both accuracy and F_1 measure in a certain iteration is the *AlgarroboNet*. These results are consistent with previous research using different convolutional networks in the identification or classification tasks of plant



(a) Accuracy



(b) F_1 -measure

FIGURE 9: Boxplot representation of the resulting distribution of (a) accuracy and (b) F_1 -measure obtained on test, for the models with different architectures

species using RGB from aerial images [55]–[60]. In this way, these results evidence the feasibility of applying the proposed methodology to the classification of *plus* and *non-plus* *Algarrobo* trees.

In order to compare the best possible model trained in the whole experimentation process, the models with the highest F_1 -measure were selected, and the performance was summarized in Table 4. In that sense, Table 4 presents the complexity of the architecture for each model, as well as the resulting performance for the iteration that achieved the highest F_1 -measure. According to Table 4, the architecture with fewer trainable parameters and, therefore, the lowest training complexity is the proposed *AlgarroboNet*. In addition, once the model is trained, the architecture with 18 layers allows for fast feedforward computation of predicted labels, compared to the other three reference models. Looking on the right side of Table 4, the model that obtained the best performance in terms of both F_1 measure and accuracy is the proposed *AlgarroboNet* at the training iteration 29, followed very closely

TABLE 4: Architecture, trainable parameters and performance of the four models at their best trial (iteration), in terms of Accuracy, Precision, Recall, and F_1 -measure. On the left side of the table, underlined numbers emphasize the smallest (less complex) model, and on the right side, bold numbers highlight the highest performance

Neural Network	Total layers	Convolutional layers	Trainable parameters	Iteration	Maximum F_1 measure	Accuracy	Precision	Recall
AlexNet	25	5	60 965 224	12	0.98282	0.98288	0.97945	0.98621
DenseNet	708	200	20 030 952	14	0.97945	0.97945	0.97279	0.98621
GoogleNet	144	59	6 998 552	10	0.97595	0.97603	0.97260	0.97931
AlgarroboNet	<u>18</u>	<u>4</u>	<u>6 606 806</u>	29	0.98630	0.98630	0.97959	0.99310

by the *AlexNet* model at the training iteration 12. It is also relevant to say that the latter is a model considerably more complex than the proposed *AlgarroboNet*. The complexity and transfer learning applied to the *AlexNet* are two elements that explain its higher stability compared to the proposed approach. However, the evidence of higher performance with lower complexity, as provided by the architecture of the *AlgarroboNet*, makes this model preferable for implementation in practice and plausible for further improvement following the path of the so-called Green Computing [61].

Further analysis of the performance of the reference and proposed models can be achieved by looking closely at the resulting *CMs*. Fig. 10 presents the *CM* corresponding to the model with the maximum performance achieved by each of the architectures under evaluation. When comparing all cases, the lowest number of $FN = 3$ was achieved by both *AlexNet* and *AlgarroboNet*. In that sense, the impact of falsely rejected *plus* trees may reduce the possibility of protecting important specimens for preserving Algarrobo trees. On the other hand, falsely accepting *non-plus* trees may lead to extra expenses to consider special facilities to protect and favor the reproduction of *non-plus* trees, which may not be the best to preserve the species. And according to Fig. 10, in these particular settings, the model that carries the lowest cost in both errors is the proposed *AlgarroboNet*.

In summary, although all the models compared provided competitive performance, the highest accuracy and F_1 -measure was achieved by one of the models trained with the proposed *AlgarroboNet* architecture, which is the one with the lowest complexity and does not require transfer learning. However, performance stability is still an issue that needs to be addressed, and future research directions may focus on training with more iterations or take advantage of strategies to adjust the hyperparameters of the network, as explored in [50]. Furthermore, it is known to work better with a larger number of training samples [32], and retrieving more samples or using common approaches for data enhancement may be advantageous.

V. CONCLUSIONS

In this paper, a methodology is proposed for massive monitoring and characterization of *plus* and *non-plus* algarrobo trees (*Neltuma pallida*), based on RGB aerial imagery and deep learning classifiers. According to the experiments, it was shown that the methodology is suitable for implemen-

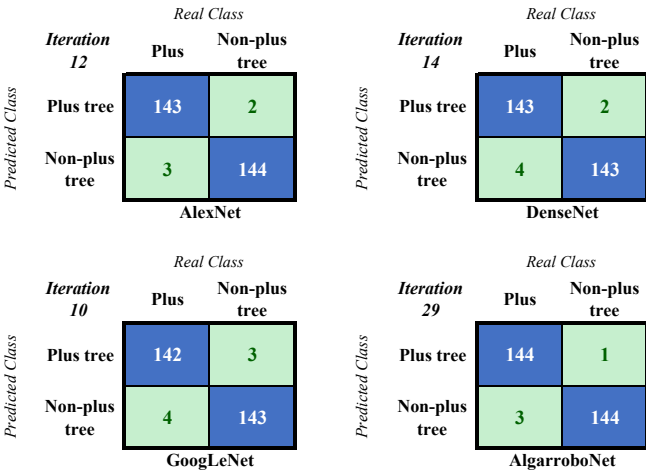


FIGURE 10: Confusion matrices *CFs* corresponding to the model that achieved the maximum F_1 -measure among the 30 experimental replicas, for the four deep classifiers that were evaluated using the proposed methodology

tation to maintain an up-to-date inventory and enforce reforestation programs. In that sense, aerial images of *plus* and *non-plus* algarrobo trees were employed to calculate morphometric characteristics for manual classification. Additionally, the proof of concept case of study was implemented with three commonly used CNN-based approaches (*AlexNet*, *DenseNet*, and *GoogleNet*) to evaluate the methodology on selected zones of interest in Peru, that the National Forest Service of Peru previously analyzed. Finally, a new application-specific architecture was proposed for efficient classification of *plus* and *non-plus* Algarrobo trees, the so-called *AlgarroboNet*. Results reveal that the proposed methodology was successfully applied with *state-of-the-art* approaches, as well as the proposed architecture. The different models present a trade-off between computational resources and accuracy.

In summary, although all the models compared provided competitive performance, the highest accuracy, and F_1 -measure was achieved by one of the models trained with the proposed *AlgarroboNet* architecture. This model results in being the one with the lowest complexity among the models compared in the case of the study, and additionally, it does not require transfer learning. However, a drawback of the *AlgarroboNet* architecture is the variability of its resulting

performance and further exploration of training strategies such as data augmentation and hyperparameter optimization should be explored.

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WILSON CASTRO (SM'23) received his Ph.D. degree in food science, technology and management from the Polytechnic University of Valencia, Spain. He is currently a Senior Researcher Professor at the Department of Food Engineering, Universidad Nacional de Frontera, Peru. He is focused on applications of images, pattern recognition, artificial intelligence, and software development to analyze agri-food products.



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HIMER AVILA-GEORGE (SM'18) received the Ph.D. degree in computer science from the Polytechnic University of Valencia, Spain. He is currently a Full Professor at the University of Guadalajara, Mexico, where he also serves as a Research Coordinator at the Centro Universitario de los Valles. He is a distinguished researcher specializing in artificial intelligence, focused on developing and applying computational techniques to address sustainability and environmental challenges.



WILLIAM NAURAY is currently a Wild Flora and Environmental Management Specialist at the Directorate of Studies and Research of the National Forestry and Wildlife Service of Peru. His research interests include flora management of dry forests and grasslands and taxonomy.



ROENFI GUERRA is an Agroindustrial engineer, M.Sc degree in Environmental Engineering, and D.Sc. in Sciences Technology and Environment. Associate professor at the Department of Agroindustrial Engineering, Universidad Nacional del Altiplano, Peru. Focused on environmental management.



JORGE RODRIGUEZ, born in Sullana, Piura, Peru. He graduated in agricultural engineering from the National University of Piura, which is part of the GICS-UNP Soil Sciences Research Group.



JORGE CASTRO, born in Trujillo, Peru, in 1983. He obtained a degree in Biology from the National University of Trujillo in 2009 and later completed a Master's Degree in Environmental Engineering at the National University of Cuyo. He is pursuing additional master's degrees in Integrated Water Resources Management and Teaching in Digital Settings, both at the National University of Cuyo.



MIGUEL DE-LA-TORRE (SM'19) earned his PhD at the École de technologie supérieure, Canada. Since 2006, he is a Full Professor at the University of Guadalajara. He has been Chair of the Guadalajara Chapter of the IEEE Geoscience and Remote Sensing Society (2023–2024). His research interests include the theory and application of machine learning and adaptive pattern recognition to address sustainability and food engineering challenges.